

Dr. Santosh Todkar

Applied AI Engineer | Bangalore|7744004068| todkarsant@gmail.com | [LinkedIn](#) | [Portfolio](#) | [GitHub](#) | [Research](#)

Summary -

Applied AI Engineer with 3.5+ years of experience building production-grade Agentic AI, GenAI, and ML solutions across BFSI, Telecom, and Industrial domains. Delivered enterprise AI products that reduced document validation time by 80%, customer churn by 33%, and improved operational efficiency by 25% using Azure OpenAI, LangGraph, and Azure AI Foundry.

Experience - (3 years 6 months)

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| 1. AI Engineer, Acronotics Limited | September 2025 to date |
| 2. Data Engineer, Reach Mobile (Datami Mobile Solutions Pvt Ltd) | February 2023 to July 2025 |
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Technical Expertise -

1. **LLM Platforms & Models** – Azure OpenAI, gpt-4.1, gpt-5, Claude Sonnet 3.5, Ollama
 2. **Retrieval & Search** – Azure AI Search (Azure Cognitive Search), FAISS, ChromaDB
 3. **Advanced RAG Technologies** – Azure AI Document Intelligence, Hybrid Retrieval, Semantic Search, Cross-Encoder Reranking (bge-reranker), Prompt & Retrieval Caching
 4. **LLM Evaluation & Responsible AI** – Azure evaluators, Groundedness, Similarity, Relevance scoring, prompt validation and trusted AI practices, RAGAS
 5. **LLM Observability** – LangSmith, Human-in-the-Loop, Hallucination Detection, Context Engineering
 6. **Statistics** – Hypothesis testing, A/B testing, Econometrics, Causal inference methods
 7. **ML Evaluations/Explainability** - Confusion matrix, accuracy, precision, recall, F1-score, AUC-ROC, MSE, MAE, adjusted R² score, SHAP and LIME.
 8. **ML orchestration tools** – AutoML, MLflow
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Enterprise GenAI & AI Automation Solutions -

Acronotics Limited - (Enterprise AI startup delivering GenAI solutions for banking and industrial clients.)

A. Banking Document Intelligence AI Validation Platform

1. Built and deployed a RAG-based document extraction and validation platform using Azure OpenAI for automated Letter of Credit document processing.
 2. Reduced Letter of Credit document extraction & validation time from 2.5 hours to 15 minutes.
 3. Designed asynchronous AI workflows using Celery and Redis for concurrent document processing, supporting a batched production pipeline of 30 documents per day.
 4. Led end-to-end delivery of AI solutions, collaborating with architects, business stakeholders, and mentoring junior engineers from solution design through production deployment.
 5. Designed prompt and retrieval strategies that improved reasoning quality, groundedness, and extraction accuracy for enterprise financial documents.
 6. Developed production-grade FastAPI microservices enabling enterprise AI integration.
 7. Implemented hybrid retrieval combining vector search, semantic ranking, and LLM evaluation to improve extraction accuracy and reduce hallucinations.
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B. AI Contract Price Intelligence Application

1. Architected a multimodal RAG system using Azure OpenAI and Azure AI Document Intelligence for intelligent contract price benchmarking across unstructured documents.
2. Implemented hybrid retrieval using vector embeddings, semantic search, and BGE Cross-Encoder (bge-reranker) to improve retrieval precision and contextual relevance.

3. Built stateful LangGraph-based agentic workflows with tool-calling capabilities and transformer-based document intelligence for multi-step reasoning, contract analysis, and GenAI-assisted price benchmarking.
 4. Optimized production inference through prompt and retrieval caching, reducing latency and LLM token consumption for enterprise workloads.
 5. Collaborated with business stakeholders, architects, and junior engineers to deliver and continuously optimize AI workflows.
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Reach Mobile – (Telecom analytics company supporting US mobile operators through predictive analytics.)

A. AI Customer Retention Platform

1. Developed an NLP-powered customer retention model that reduced churn by 33%, enabling more targeted retention campaigns and improving customer lifetime value.
 2. ML pipeline deployment - Automated NLP feature engineering using topic modelling, sentiment analysis, and keyword extraction with real time pipeline.
 3. Model selection/ML Model ranking - eXtreme Gradient Boosting (best performance after comparison with Random Forest, Logistic Regression)
 4. Drift detection: Monitored model performance monthly, retrained if ROC-AUC dropped below 0.80.
 5. SHAP and LIME validation techniques were used for the model evaluation and explainability.
 6. Delivered predictive analytics solutions supporting customer retention and operational decision-making.
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B. AI Inventory Forecasting Platform

1. Built an XGBoost-powered forecasting platform that improved inventory replenishment efficiency by 25% through automated stock prediction and threshold-based alerts.
 2. Conducted exploratory data analysis, feature engineering, and data validation to improve predictive model performance and support production deployment.
 3. ML pipeline deployment - Data collection from AWS S3 bucket, DynamoDB, preprocessing, training, validation, and deployment
 4. Showed predicted V/s Actual SIM stock over time - eXtreme Gradient Boosting regression
 5. Inventory alert system – Highlights low stock or high stock based on thresholds and suggested EOQ for the replenishment
 6. Drift detection: Tracked model refresh cycles, last retrain date, and performance drift
 7. Collaborated with cross-functional teams and mentored interns to translate analytics solutions into measurable business outcomes.
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Education – (Shivaji University, Kolhapur, Maharashtra)

1. PhD- (July 2018-Sept. 2024)
 2. M.E-Mechanical Design (2010-2012)
 3. B.E- Mechanical (2005-2009)
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Research & Publications –

1. PhD (Mechanical) - Composite materials and sustainable biocomposites.
 2. Research direction: ML-assisted material characterization, surrogate modelling and multi-objective optimization for sustainable engineering materials.
 3. Citations 300+, Published the papers in peer reviewed journal, impact factor – 12, Q1 journals
 4. Publications – Elsevier journal – Composites Part : B
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AI Certifications –

1. Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate.
 2. AMD AI Developer Academy - Reinforcement Learning with Large Language Models.
 3. Introduction to Claude code, Claude cowork, Claude 101
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